OncoBrief V4 -- Master Enhancement Plan & Architectural Blueprint

Excelling Across Use Case 02 (Consultation Readiness), Use Case 04 (Clinic Operations & Patient Flow), Use Case 05 (Doctor Productivity & Knowledge Assistant), and All Hackathon Design Principles

* Target Repository: /Users/dhaanish/Documents/OncoBrief
* Target Tracks: Cancer Care (National Cancer Grid / NCG-KCDO), Maternal & Child Health (FOGSI), and Diabetes Care (RSSDI.pdf))
* Standards & Stack: TypeScript Monorepo (pnpm@9.15.4), Next.js 15 App Router, PostgreSQL 16 + Forced RLS, NRCeS ABDM v6.5.0, HL7 mCODE STU4, Sarvam AI (doc-digitise, saaras:v3, bulbul:v3), Amazon Bedrock Nova Pro, and Deterministic Agentic State-Graph Orchestration

1. Executive Charter, Vision & Target OKRs

1.1 Overarching Vision

OncoBrief V1-V3 established a strong foundation: an append-only, span-anchored Evidence Ledger (packages/db/src/services/ledger.ts), a two-stage deterministic provenance gate (packages/domain/src/provenance/span-validation.ts), a side-by-side Reconciliation Room (apps/web/src/app/(app)/patients/[id]/conflicts/[conflictId]/page.tsx), and strict code-level enforcement of the non-clinical boundary (packages/domain/src/policy/clinical-boundary.ts).

However, our audit against the official hackathon rubric revealed that OncoBrief currently operates as a single-track, high-friction new-patient document intake ledger rather than a low-click, longitudinal clinical operations and productivity platform:

1. Use Case 02 (Consultation Readiness & Patient Journey Review) Gaps: It lacks a "What Changed Since Last Visit" interval delta view, misclassifies longitudinal serial lab trends as cross-document contradictions, drops documented pending investigation instructions, hardcodes oncology-only UI labels, and bypasses cloud/Indic OCR on local web uploads.
2. Use Case 04 (Clinic Operations & Patient Flow) Gaps: It has zero scheduling, slot, queue, arrival, walk-in, or waitlist tables, leaves tasks permanently "unassigned" in the UI, and never uses its readiness_band to prevent day-of-clinic bottlenecks or backfill cancelled OPD slots.
3. Use Case 05 (Doctor Productivity & Knowledge Assistant) Gaps: Its assistant classifier blocks all guideline and protocol lookups, has no institutional knowledge corpus, and refuses to draft audience-adapted professional artifacts (referral letters, tumor-board decks, academic case reports, or patient handouts).
4. Design Principles & Accessibility Gaps: It requires 45+ manual clicks to verify a single patient's chart (no batch verify), has zero caregiver data model or spoken voice-note preview on the Care Continuity page, leaves browser extension and ABDM/mCODE imports as alerts/stubs, and suffers from 375px mobile overflow and WCAG 2.2 AA contrast/label defects.

The V4 Strategic Upgrade transforms OncoBrief into a unified, multi-use-case Clinical Readiness, OPD Flow & Grounded Knowledge Platform where:

* Every follow-up consultation opens with an instant "Changes Since Last Visit" Delta Brief, longitudinal lab/medication flowsheets, and 1-click batch verification across Oncology (NCG), Diabetes (RSSDI), and Maternal Health (FOGSI) template packs (Use Case 02).
* Every outpatient clinic runs a Readiness-Gated Smart OPD Queue that tracks arrivals, walk-ins, delayed investigations, and clinician-assigned administrative priority tiers--automatically backfilling cancelled or blocked slots with waitlisted patients whose consultation packets are already verified ready (Use Case 04).
* Every clinician has a Source-Grounded Productivity & Protocol Co-Pilot that searches span-anchored institutional SOPs and society guidelines (NCG, FOGSI, RSSDI) with bounding-box provenance, and drafts editable referral letters, tumor-board slide outlines, academic case summaries, and multilingual patient handouts where every sentence cites [E1]/[G1] ledger spans (Use Case 05).
* Every patient and family caregiver receives verified administrative reminders and pre-procedure checklists as playable regional-language voice notes (Sarvam Bulbul v3 across 11 Indic languages) with two-way attendance confirmation (Design Principles & Use Case 01).

1.2 Quantitative Objectives & Key Results (60-90 Day Pilot OKRs)

Objective	Key Result (Self-Measured in Database Telemetry)	Baseline (V3)	Target (V4 Pilot)

O1: Excel at Consultation Readiness & Longitudinal Review (Use Case 02)	KR1.1: Reduce clinician clicks required to verify a 9-document referral packet via confidence-tiered batch verification (verifyEvidenceBatchAction).	45+ clicks	<= 4 clicks
	KR1.2: Eliminate false-positive conflicts on longitudinal serial labs and medication adjustments across different dates (observed_on).	100% false-conflict rate on serial labs	0% false conflicts on dated series
	KR1.3: Support complete longitudinal template packs across all 3 tracks (NCG Oncology, FOGSI Maternal Health, RSSDI Diabetes).	1 track (Oncology only)	3 tracks seeded & configurable
O2: Optimize OPD Clinic Flow & Slot Utilization (Use Case 04)	KR2.1: Track live OPD slot states (scheduled, checked_in, in_consult, delayed_investigation, cancelled, walk_in) and recommend ready waitlist backfills in < 5 seconds.	0 slot/queue tracking	100% slot state coverage + automated backfill
	KR2.2: Achieve 100% task ownership assignment across coordinators, nurses, and records staff via in-row assignee controls.	0% UI-assigned tasks (unassigned)	100% assignable in 1 click
O3: Empower Doctor Productivity & Knowledge Synthesis (Use Case 05)	KR3.1: Deliver 100% span-grounded answers over uploaded institutional SOPs and NCG/FOGSI/RSSDI guideline PDFs ([G1] citations with bbox preview).	0% (all guideline queries refused)	100% grounded SOP/guideline retrieval
	KR3.2: Generate 4 clinician-editable professional document formats (Referral Letter, Tumor Board Deck, Case Report Draft, Patient Handout) with 0 unanchored claims.	Raw JSON export only	4 structured, source-cited formats
O4: Meet Every Human-Centered, Multilingual & Safety Principle	KR4.1: Enable caregiver registration and playable 11-language spoken audio reminders (Sarvam Bulbul v3) on /continuity.	Text-only En/Hi templates; 0 caregiver fields	11 Indic languages + playable TTS audio + caregiver sync
	KR4.2: Achieve 100% WCAG 2.2 AA contrast (>= 4.5:1 text, >= 3.0:1 UI borders), 0 unlabeled form inputs, and 0 px horizontal overflow at 375px mobile width.	135+ unlabeled inputs; +99px to +266px mobile overflow	0 axe/a11y violations; 0px mobile overflow

1.3 Explicit Non-Goals & Strict Out-of-Scope Boundaries

To preserve regulatory safety and 100% compliance with the hackathon's Out of Scope rules (packages/domain/src/policy/clinical-boundary.ts), OncoBrief V4 will NEVER:

1. Compute or infer a medical diagnosis, cancer stage (TNM), or RECIST tumor response--it only transcribes documented values verbatim (diagnosis_text.as_written, stage_text.as_written) and retrieves verbatim passages from human-approved institutional guideline documents.
2. Generate autonomous treatment recommendations or clinical decision support (CDS)--when searching institutional SOPs or NCG/FOGSI/RSSDI guidelines, the assistant retrieves and quotes the exact published guideline section with its source bounding box so the clinician reads and applies it; the system never tells the doctor what drug or dose to prescribe for a patient.
3. Calculate a clinical risk score, physiological acuity score, or autonomous medical triage--OPD queue prioritization in Use Case 04 uses clinician-assigned or referring-doctor-documented administrative priority tags (stat_referral_documented, post_op_check_documented, expedited_slot_requested, standard_followup) anchored to a source document or explicit human selection, never an AI-computed medical severity score.

4. Interpret medical images (DICOM pixel arrays), pathology slides, or genomic variants--multi-modal adapters extract only administrative headers, stated report text, and tabular lab rows as written.
5. Execute silent automation--every batch verification, conflict resolution, slot backfill offer, referral letter export, and patient/caregiver message requires explicit human confirmation and appends a tamper-evident SHA-256 hash-chained audit event.

2. Deep Competitive & Ecosystem Research

2.1 Cross-Use-Case Competitive Landscape Matrix

Platform / System	Use Case Coverage	Core Strengths in Production	Critical Gaps in Indian / NCG Context	How OncoBrief V4 Beats or Bridges It
Vizlitics Consult Prep (Cancer Insights)	UC 02 (Oncology Consult Prep)	Pulls electronic records from 93% of US health systems; builds source-linked longitudinal cancer summaries and trial matches.	Assumes US HIE interoperability; relies on generative clinical summarization rather than deterministic fact-state separation (extracted vs verified) or conflict reconciliation rooms.	Adds zero-integration multi-report scan segmentation, deterministic two-stage span entailment, explicit contradiction reconciliation, and NRCeS ABDM v6.5.0 export.
GE HealthCare CareIntellect for Oncology	UC 02 (Longitudinal Timeline)	Summarizes multi-modal hospital data (DICOM, labs, notes) into a single view and flags progression since prior visits.	Enterprise hospital-locked deployment; crosses into clinical progression interpretation; no outpatient slot backfill or multilingual caregiver outreach.	Adopts CareIntellect's "Changes Since Last Visit" delta banner and serial lab trend flowsheets while keeping every value strictly .as_written and non-interpretive.
Navina Clinician Copilot (Core Tech)	UC 02 (Pre-Visit Chart Review across Chronic Care)	Best-in-class problem-oriented pre-visit summary across diabetes, cardiology, and primary care; links claims to PDF bounding boxes.	Focused on US Value-Based Care (HCC coding/RAF scores); lacks Indian regional language OCR/voice and OPD queue/slot management.	Adopts Navina's chronic-care versatility (adding Diabetes & Maternal Health packs) and interval-change cards while avoiding diagnostic HCC coding.
Tempus One & Tempus Next	UC 02 & UC 05 (Oncology AI & Care Gaps)	LLM timeline construction from unstructured PDFs (ASCO 245826) and real-time guideline gap alerts.	Operates as a clinical decision support (CDS) and molecular profiling platform (out of scope for non-clinical guardrails).	Keeps guideline retrieval strictly as a clinician-initiated, span-grounded reference library ([G1]) rather than autonomous clinical alerting.
Honey Health (Pre-Charting)	UC 02 & UC 04 (Referral Intake & Pre-Charting)	Autonomous AI workers ingest referrals, verify insurance authorizations, and pre-chart interval ER/consult notes before the visit.	US EHR-specific; lacks cryptographic append-only audit chains and side-by-side source contradiction resolution.	Adopts Honey Health's low-click pre-charting ergonomics (1-click batch verification of high-confidence spans and automated pending-order task extraction).
OpenEvidence (PMC Study)	UC 05 (Medical Literature & Guideline Co-Pilot)	Searches peer-reviewed literature and society guidelines (NCCN, AAP) to synthesize cited answers for teaching, research, and practice.	Queries external global literature without awareness of a hospital's local SOPs or the patient's administrative record readiness.	Adds an Institutional SOP & Guideline Ledger (guideline_document + guideline_span) so doctors can query NCG/FOGSI/RSSDI protocols alongside patient records with exact bbox provenance.
Doximity GPT & Glass Health	UC 05 (Clinical Writing & Admin Productivity)	Drafts referral letters, prior-auth appeals, after-visit summaries, and reading-level-adapted patient education handouts in seconds.	Prompt-only generation without character-level substring validation against underlying source PDFs--vulnerable to subtle date/lab hallucinations.	Builds an Audience-Adaptive Studio where referral letters, tumor-board decks, and patient handouts are compiled exclusively from verified ledger facts ([E1]).

LeanTaaS iQueue & Phreesia	UC 04 (OPD Flow, Scheduling & Slot Backfill)	LeanTaaS optimizes oncology clinic/infusion templates across 36+ NCI centers; Phreesia Appointment Accelerator backfills cancelled slots in ~5 minutes.	Neither system inspects whether a waitlisted patient's outside medical records, pathology slides, and insurance auths are actually ready before slotting them in.	Introduces Readiness-Gated Slot Backfill: when an OPD slot opens, OncoBrief recommends waitlisted patients whose record readiness band is already ready.
Qventus & Eka Care	UC 04 & UC 02 (Pre-Visit Clearance & Indian OPD Queue)	Qventus flags missing pre-op clearances before surgery; Eka Care manages Indian OPD token queues and ABHA ID linkage.	Eka Care lacks multi-document contradiction reconciliation; Qventus is built for US inpatient/perioperative suites.	Combines Eka Care's Indian OPD token/walk-in queue and ABDM integration with Qventus-style pre-visit readiness gating and Sarvam Indic voice outreach.
Navya Care & Karkinos Healthcare	Indian Oncology Ecosystem	Navya provides NCG/TMC expert treatment opinions; Karkinos provides KareMitra navigators and a Command Centre.	Rely heavily on manual human chart abstraction to assemble unstructured paper/WhatsApp records before tumor-board review.	Serves as the upstream automated Record-Readiness & Provenance Engine that feeds clean, verified packets into NCG tumor boards and navigation workflows.

2.2 Multi-Track Clinical Domain Research (Cancer, Diabetes & Maternal Health)

Use Case 02 explicitly requires supporting long-term care settings such as cancer, diabetes, and maternal health, where every consultation builds upon previous decisions:

1. Cancer Care (NCG-KCDO Oncology EMR Initiative): Requires tracking multi-modality episodes (histopathology/IHC receptor status, surgical operative notes, chemotherapy cycle administration logs, radiation completion summaries, serial tumor markers/CBCs, and insurance pre-authorizations like AB-PMJAY) across 380+ NCG hospitals.
2. Maternal & Child Health (FOGSI-ICOG Good Clinical Practice Recommendations): Requires milestone-driven longitudinal tracking across trimesters: Last Menstrual Period (LMP) & Estimated Due Date (EDD), serial Antenatal Care (ANC) visit blood pressure and weight logs, serial Hemoglobin and Oral Glucose Tolerance Test (OGTT) results, TIFFA/anomaly ultrasound scans, Tetanus/Tdap immunization dates, and post-partum follow-up checklists.
3. Diabetes Care (RSSDI Clinical Practice Recommendations.pdf)): Requires decade-spanning longitudinal flowsheets of Fasting/Post-Prandial Blood Glucose (FBG/PPBG), serial HbA1c trajectories, renal function panels (Serum Creatinine, eGFR, Urine Albumin-to-Creatinine Ratio / UACR), lipid profiles, annual dilated retinal/fundus exam dates, peripheral neuropathy foot screening records, and oral hypoglycemic / insulin dose adjustments.

3. Root-Cause Summary of Current Application Failures

Module / Subsystem	Exact File & Line in Repo	Current Failure Mode	Impacted Requirement
Serial Lab & Medication Conflict Comparator	packages/domain/src/vocab/fact-types.ts:103-147 & packages/domain/src/conflict/comparators.ts:144-169	slotQualifier groups all lab_result.as_written by lab:<name> and all medication.recorded by drug:<name> regardless of observed_on date. Two CBC reports from June and July with different Hemoglobin values are flagged as a contradiction (disagree).	UC 02: Longitudinal lab trends & medication history
Missing "Changes Since Last Visit" Delta Engine	apps/web/src/app/(app)/patients/[id]/evidence/page.tsx:50-56 & packages/db/src/services/patients.ts:95-153	getTimeline returns a flat list grouped by YYYY-MM. No concept of last_consultation_date or comparison against the previous approved consultation_packet exists.	UC 02: Quickly understand what changed since last visit

Ignored Pending Investigation & Action Instructions	packages/domain/src/vocab/fact-types.ts:13-31 & packages/adapters/src/rule-based-extractor.ts:59-250	FACT_TYPES has no entry for pending investigations or explicit clinical-admin instructions (e.g., "Bring prior imaging and histopathology slides", "Awaiting IHC"), so RuleBasedExtractor drops them.	UC 02 & UC 04: Pending investigations & action queues
Hardcoded Local Upload Adapters & Stubbed Image OCR	apps/web/src/lib/actions.ts:313-320 & packages/adapters/src/local-multimodal-ocr.ts:387-436	uploadDocumentAction hardcodes LocalMultimodalOcrAdapter + RuleBasedExtractor (ignoring OCR_DRIVER); extractImage only returns image dimensions and SHA-256 without OCRing text in photos; SarvamVisionOcrAdapter.splitPdfIntoBatches (sarvam-vision-ocr.ts:98-118) replaces >10-page PDFs with synthetic stub text.	UC 02 & Design Principles: Multi-modal photo/scan OCR
Fake EHR/ABDM Import & Link Tabs	apps/web/src/components/DocumentUploadForm.tsx:320-374	"Import from EHR / ABDM" buttons only call browser alert(), and "Link to Existing" search button has no handler.	UC 02 & Workflow First Principle
Hardcoded Oncology Labels in Timeline View	apps/web/src/components/TimelineNodeView.tsx:48,138	Hardcodes 'Apollo Hospitals' when issuingFacility is null and labels every imaging fact 'CT Scan (Thorax)'; cannot represent Diabetes or Maternal Health modalities.	UC 02: Cancer, Diabetes & Maternal Health generality
Zero OPD Scheduling, Slot, Arrival, or Waitlist Engine	packages/db/migrations/0002_schema.sql & apps/web/src/app/(app)/workspace/page.tsx:31-90	No tables for appointment_slot, opd_queue_entry, or waitlist_entry. Workspace table does not even display appointmentDate, arrival status, walk-ins, or slot backfill options.	UC 04: Clinic Operations & Patient Flow
Unassigned Tasks & Dead Server Actions	apps/web/src/app/(app)/patients/[id]/tasks/page.tsx:73-114 & apps/web/src/lib/actions.ts:186,223,330	assignTaskAction, waiveGapAction, and withdrawPacketAction have no UI triggers; all tasks created via UI render "unassigned".	UC 04 & UC 02: Care-team coordination
Assistant Blocks All Guideline/SOP Queries & Content Drafting	apps/web/src/lib/assistant/classify.ts:30-54, tools.ts:301 & packages/adapters/src/bedrock-llm.ts:308-311	classify.ts blocks all medical/guideline questions; no institutional SOP/guideline table exists; PACKET_SYSTEM_PROMPT forbids prose drafting (referral letters, slide outlines, case reports, handouts); getCaseContext (tools.ts:301) computes openTasksCount from timeline instead of listTasks.	UC 05: Doctor Productivity & Knowledge Assistant
High Click Burden, Missing Caregiver Model & A11y/Mobile Bugs	apps/web/src/components/EvidenceRow.tsx, continuity/page.tsx, audit/page.tsx	No batch verification UI (45+ clicks/patient); no caregiver_contact or TTS audio preview on /continuity; state-extracted (#9a6b12) and rule-strong (#b9b0a0) fail WCAG AA contrast; horizontal overflow at 375px mobile width.	Design Principles: Easy for doctors/patients, Multilingual & Caregiver-friendly

4. Functional Specifications & User Stories (Given-When-Then)

Epic 1: Longitudinal Patient Journey, Interval Deltas & Multi-Track Readiness (Use Case 02)

Story 1.1: "Changes Since Last Consultation" Interval Delta Brief

* User Story: As a consulting clinician preparing for a follow-up visit, I want to see a dedicated "Changes Since Last Consultation" banner and filter on the Evidence Journey page so that I can immediately identify new documents, new lab/imaging results, reopened conflicts, and pending tasks since the patient's last encounter without re-reading the entire historical chart.
* Acceptance Criteria (Gherkin):
  * Scenario 1: Patient has a prior consultation date or approved packet
    * Given a patient record has last_consultation_date = '2025-07-05' (or a prior approved consultation_packet approved at ledger_seq = 18)
    * When the clinician opens /patients/[id]/evidence
    * Then the top of the page renders a "Changes Since Last Visit (05 Jul 2025)" summary card displaying exact counts and 1-click filter pills for: New Documents (+3), New / Updated Facts (+8), New / Reopened Conflicts (+1), and Pending Investigations / Tasks (+2)
    * And clicking the Since Last Visit filter chip filters both the Timeline View and Ledger Details View to display only facts with observed_on > '2025-07-05' or ledger_seq > 18, badged with a high-contrast NEW SINCE LAST VISIT indicator.

Story 1.2: Longitudinal Serial Lab Flowsheets & Medication Regimen History (Fixing False Conflicts)

* User Story: As a clinician reviewing a chronic care patient (oncology, diabetes, or maternal health), I want serial laboratory results (e.g., Hemoglobin, HbA1c, Serum Creatinine, Platelets) and dated medication records across different dates to be grouped into a Longitudinal Flowsheet & Regimen Timeline rather than flagged as contradictions, while still flagging genuine same-date cross-document disagreements as conflicts.
* Acceptance Criteria (Gherkin):
  * Scenario 1: Serial labs on different dates form a trend series, not a conflict
    * Given Document A (2025-06-02) records Hemoglobin: 11.4 g/dL and Document B (2025-07-26) records Hemoglobin: 10.2 g/dL
    * When detectAndRecordConflicts runs during ingestion
    * Then slotQualifier scopes lab and medication comparison by date (lab:hemoglobin@2025-06-02 vs lab:hemoglobin@2025-07-26) or COMPARATORS['lab_result.as_written'] returns 'agree' (non-conflicting series) when observedOn dates differ by > 1 day
    * And the Evidence Journey page renders a "Longitudinal Lab & Vitals Flowsheet" table showing Test Name | Prior Value (Date) | Latest Value (Date) | Delta (As Written) | Source Links with zero clinical interpretation.
  * Scenario 2: Same-date cross-document lab disagreement triggers a conflict
    * Given two documents both dated 2025-07-26 record Hemoglobin: 10.2 g/dL and Hemoglobin: 13.8 g/dL
    * When detectAndRecordConflicts runs
    * Then both facts are marked conflicting and surfaced in the Reconciliation Room.

Story 1.3: Automated Extraction of Pending Investigations & Documented Instructions

* User Story: As a care coordinator, I want OncoBrief to extract explicitly documented pending investigations (e.g., "Awaiting IHC report", "Pending PET-CT", "OGTT scheduled") and patient instructions (e.g., "Bring prior imaging CD and histopathology slides at next visit") as span-anchored facts that can be promoted into administrative tasks in one click.
* Acceptance Criteria (Gherkin):
  * Given a discharge summary contains "Bring prior imaging and the histopathology slides at the next visit"
  * When the document is ingested via RuleBasedExtractor or BedrockLlmProvider
  * Then the extractor emits a candidate with factType: 'instruction.recorded' (or 'pending_investigation.recorded') anchored to the exact span
  * And the Evidence Row and Record Map display a Create Task from Instruction button pre-filled with the verbatim instruction and source link.

Story 1.4: Multi-Track Chronic Care Template Packs (NCG Oncology, FOGSI Maternal Health, RSSDI Diabetes)

* User Story: As a hospital administrator or clinician operating across oncology, obstetrics, or endocrinology OPDs, I want to assign domain-specific Checklist Template Packs (ncg_oncology_intake, fogsi_antenatal_care, rssdi_diabetes_followup) so that record readiness and timeline modalities reflect the patient's care track.
* Acceptance Criteria (Gherkin):
  * Given checklist_template is seeded with 3 track templates (ncg_oncology_intake, fogsi_antenatal_care, rssdi_diabetes_followup) and 5 demo patients (DEMO-001 to DEMO-003 for Oncology, DEMO-MAT-004 for High-Risk Antenatal Care, DEMO-DIA-005 for Long-Term Type 2 Diabetes)
  * When a clinician filters /workspace by Care Track or opens DEMO-MAT-004 / DEMO-DIA-005

* Then RecordMapPage evaluates track-specific required documents (e.g., anc_card, tiffa_ultrasound_report, ogtt_lab_report for Maternal Health; hba1c_lab_report, renal_panel_report, retinal_fundus_report, foot_neuropathy_chart for Diabetes)
* And TimelineNodeView dynamically assigns modality labels and icons from the actual factType and documentType without ever falling back to hardcoded 'Apollo Hospitals' or 'CT Scan (Thorax)'.

Epic 2: Readiness-Gated OPD Clinic Operations & Patient Flow (Use Case 04)

Story 2.1: Live OPD Queue, Slot Status & Walk-In / Cancellation Management

* User Story: As an OPD coordinator or front-desk manager, I want /workspace to provide a real-time OPD Clinic Flow & Slot Board alongside Record Readiness so that I can manage scheduled appointment slots, patient check-ins, walk-ins, delayed investigations, and last-minute cancellations in one view.
* Acceptance Criteria (Gherkin):
  * Given today's OPD clinic has scheduled time slots (09:00, 09:30, 10:00, 10:30, 11:00) across consulting rooms
  * When the coordinator opens /workspace (or /workspace/flow)
  * Then each slot displays: Slot Time, Patient & Token #, Attendance Status (scheduled | checked_in | in_consult | delayed_investigation | cancelled | no_show | walk_in), Record Readiness Band (ready | gaps | blocked), Documented Priority Tag (source-anchored, e.g., post_op_review / chemo_clearance / routine), and Bottleneck Alert
  * And the coordinator can update slot status (Check In, Mark Delayed Lab, Cancel Slot, Register Walk-In) with full audit logging.

Story 2.2: Readiness-Gated Vacant Slot Backfill Engine

* User Story: As an OPD coordinator, when a scheduled patient cancels last-minute or is held in the waiting room due to a delayed investigation (blocked), I want OncoBrief to automatically identify waitlisted or later-slotted patients whose consultation packets are already ready (readiness_band = 'ready') and send a human-approved slot-offer message so clinic capacity is never wasted.
* Acceptance Criteria (Gherkin):
  * Given the 09:30 slot for DEMO-001 is marked cancelled or delayed_investigation (readiness_band = 'blocked')
  * When the Smart Slot Backfill drawer opens
  * Then the system ranks candidate patients on the waitlist or later slots strictly by operational criteria: (1) readiness_band === 'ready', (2) documented administrative priority tag, and (3) earliest requested date
  * And clicking "Offer 09:30 Slot & Reassign" swaps the slot assignment, logs an opd.slot_backfilled audit event, and queues a verified appointment update message for clinician/coordinator dispatch.

Story 2.3: In-Row Care-Team Task Assignment & Escalation

* User Story: As a care coordinator, I want to assign and reassign any administrative task on /patients/[id]/tasks and /patients/[id]/record-map to a specific team member (Clinician, Coordinator, Records Officer) with a target due time so that every clinic bottleneck has a named owner.
* Acceptance Criteria (Gherkin):
  * Given an open task exists on /patients/[id]/tasks
  * When a user with task:assign permission selects a staff member from the in-row Assignee dropdown and clicks Assign
  * Then assignTaskAction updates admin_task.assigned_to, appends a task_event (action = 'assigned'), and renders assigned to <Name> (<Role>) immediately.

Epic 3: Source-Grounded Doctor Productivity & Knowledge Assistant (Use Case 05)

Story 3.1: Span-Anchored Institutional SOP & Clinical Guideline RAG ([G1] Citations)

* User Story: As a doctor preparing for a consultation, teaching session, or tumor board, I want to ask Ask OncoBrief questions about institutional clinical protocols and society guidelines (NCG Oncology, FOGSI Maternal Health, RSSDI Diabetes) and receive answers grounded strictly in uploaded, version-pinned guideline PDFs with exact page and bounding-box citations ([G1], [G2]), while keeping patient-specific treatment decisions in my hands.
* Acceptance Criteria (Gherkin):
  * Given the hospital's library contains version-pinned guideline documents (NCG Breast Cancer Management

Guidelines v2026, FOGSI Routine Antenatal Care GCPR, RSSDI Type 2 Diabetes Practice Recommendations) stored in guideline_document and guideline_span
* When a clinician asks Ask OncoBrief: "What does the NCG guideline state for post-MRM adjuvant radiation workup?" or "What is the FOGSI antenatal screening schedule for gestational diabetes?"
* Then classifyIntent routes the query to 'guideline_knowledge_question' (distinguishing general guideline lookup from autonomous patient prescribing)
* And the assistant returns a grounded synthesis citing [G1], [G2] with clickable SourceCard links that open the exact guideline page and highlight the verbatim paragraph in SourcePaneView
* And every response includes the explicit header: Reference Guideline Extract -- For Clinician Review Only (Not a Patient-Specific Order).

Story 3.2: Audience-Adaptive Professional Content & Academic Studio (/patients/[id]/studio)

* User Story: As a clinician, I want to transform a patient's verified Evidence Ledger and linked institutional guidelines into editable, source-cited professional drafts--including (1) Specialist Referral / Transfer Letter, (2) Multidisciplinary Tumor Board / Case Presentation (Markdown & Printable Slide View), (3) De-Identified Academic Case Report Abstract, and (4) Multilingual Patient & Caregiver Information Sheet--while remaining in 100% control of editing and approving the final output.
* Acceptance Criteria (Gherkin):
  * Given a patient has verified facts in the Evidence Ledger
  * When the clinician opens /patients/[id]/packet (or the new Productivity Studio section) and selects one of the 4 artifact templates (referral_letter, tumor_board_deck, academic_case_abstract, patient_info_sheet)
  * Then the deterministic/LLM composer generates a structured draft where every factual statement carries an inline [E#] provenance badge linked to its evidence_fact_id (and automatically strips patient identifiers MRN/ABHA/Name when academic_case_abstract is selected)
  * And any unverified fact is visually badged [UNVERIFIED -- Confirm Source Before Sign-Off]
  * And the clinician can edit the draft inline, sign off (packet:approve), and export as Print-Ready PDF, Markdown/Slides, ABDM v6.5.0 JSON, or HL7 mCODE STU4 JSON.

Epic 4: Human-Centered UX, Caregiver Voice Continuity & Workflow-First Integration (Design Principles)

Story 4.1: Guided "Next Step" Readiness Banner & 1-Click Batch Verification

* User Story: As a busy outpatient doctor or records officer, I want a top "Next Step to Consult Ready" action bar and a "Verify All High-Confidence Facts (OCR >= 90%)" batch action with a single confirmation modal so I can prepare a clean chart in under 60 seconds.
* Acceptance Criteria (Gherkin):
  * Given a patient has 18 extracted facts with confidence_band = 'high' (ocr_confidence >= 0.90) and state = 'extracted'
  * When the clinician clicks Review & Verify High-Confidence Batch (18) and confirms in the modal
  * Then verifyEvidenceBatchAction transitions all 18 facts to verified in one atomic PostgreSQL transaction, records individual ledger_entry rows plus a batch audit_event, and recomputes readiness_band.

Story 4.2: Consented Caregiver Network & Playable 11-Language Voice Reminders (Sarvam Bulbul v3)

* User Story: As a care coordinator supporting low-literacy patients and family caregivers, I want to register a consented family caregiver (caregiver_name, relationship, phone, preferred_locale), compose reminders in all 11 supported Indic languages, and generate a playable spoken audio voice note (Sarvam Bulbul v3 TTS) on /patients/[id]/continuity.
* Acceptance Criteria (Gherkin):
  * Given a patient has a verified appointment fact and a registered caregiver with preferred_locale = 'ta-IN' (or 'hi-IN', 'te-IN', 'bn-IN', 'mr-IN', 'kn-IN', 'ml-IN', 'gu-IN', 'pa-IN', 'or-IN')
  * When the coordinator composes a continuity reminder and clicks "Generate & Preview Spoken Voice Note"
  * Then the page calls /api/v1/assistant/voice/tts (or an offline Web Speech / WAV fallback when SARVAM_API_KEY is unconfigured) and renders an inline <audio controls> player so the care team or judge can listen to the localized voice reminder and dispatch it to both patient and caregiver.

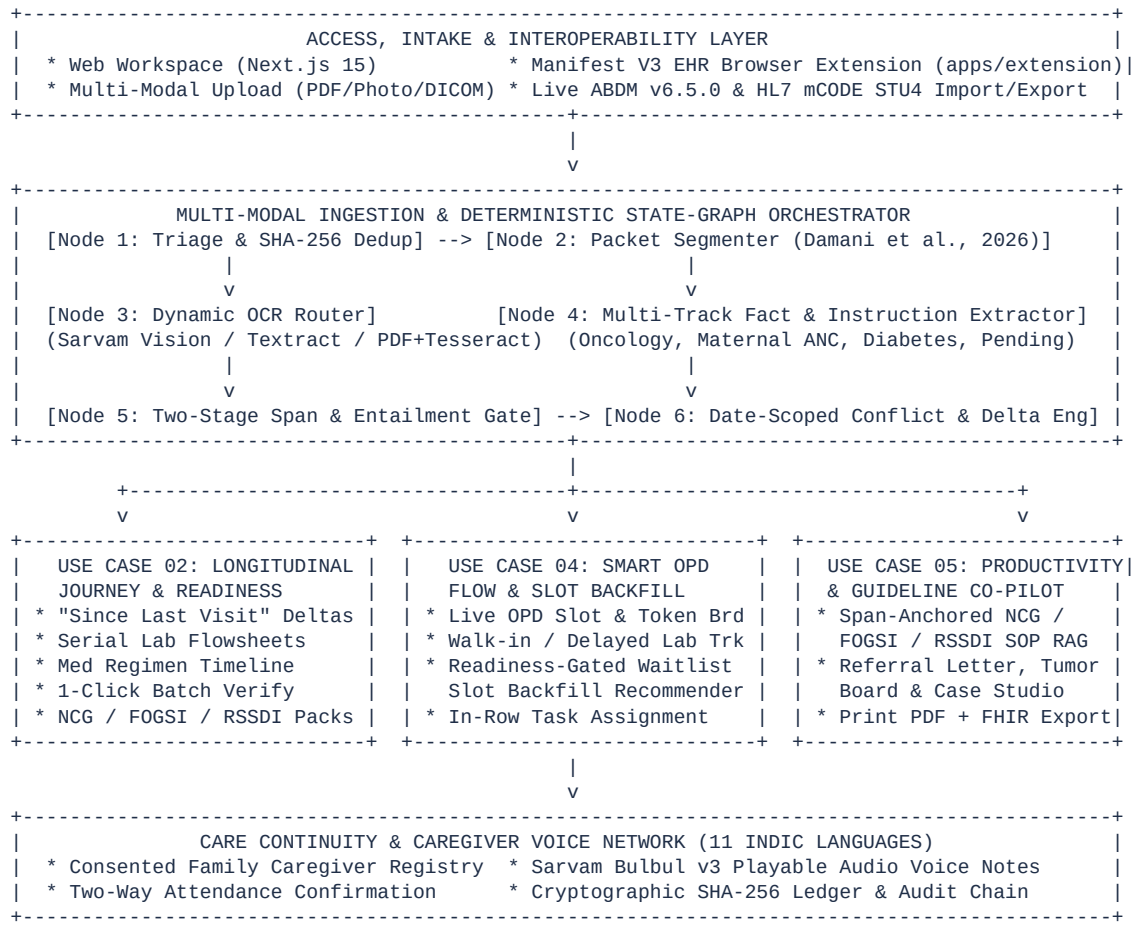
Story 4.3: Working ABDM / mCODE Intake & Browser Extension Launch Overlay

* User Story: As a clinician in an existing hospital EHR, I want the Import from EHR / ABDM and Link to Existing tabs in DocumentUploadForm.tsx to execute real server actions (ingesting valid NRCeS ABDM v6.5.0 and HL7 mCODE

STU4 bundles into the ledger), and I want a working Manifest V3 Browser Extension (apps/extension) that launches OncoBrief from any hospital EHR page using short-lived extension_grant tokens.
* Acceptance Criteria (Gherkin):
  * Given a user clicks Pull ABDM Bundle (Tata Memorial / ACTREC) or Import mCODE FHIR Bundle (AIIMS) on /patients/[id]/sources
  * When importStandardBundleAction executes
  * Then it ingests a complete, standards-valid FHIR R4 JSON bundle through createDocument + ingestDocument, extracts structured spans and facts into the Evidence Ledger, and refreshes /patients/[id]/sources with zero alert() popups.

---

5. System Architecture & Agentic Orchestration Upgrade



Why a Deterministic State-Graph Orchestrator (packages/domain/src/orchestration/workflow-graph.ts) Beats Fragile Prompt Chains

To satisfy the Technology Approach principle ("Leverage existing AI capabilities, including AI agents, orchestration frameworks...") without violating the Safety Expectations principle ("No autonomous clinical advice; human override on every step"), OncoBrief V4 implements a typed, LangGraph-compatible State-Graph Workflow Orchestrator (WorkflowStateGraph<TState>) with explicit nodes, conditional edges, checkpoint persistence, and human-in-the-loop interrupt gates:

1. Document Ingestion Graph: triageNode -> segmentPacketNode -> ocrRouteNode (selecting SarvamVisionOcrAdapter, TextractOcrAdapter, or LocalMultimodalOcrAdapter based on env.ocrDriver and document script) -> extractCandidatesNode -> twoStageSpanGateNode -> dateScopedConflictAndDeltaNode -> humanReviewInterruptGate.
2. Assistant & Co-Pilot Graph: intentClassifierNode -> branches deterministically to patientEvidenceRetrieverNode ([E#]), institutionalGuidelineRetrieverNode ([G#]), opdFlowStatusNode, or boundaryRefusalNode -> groundedSynthesisNode -> citationVerificationGateNode.

---

6. Database Schema Migration Specification (0009_v4_multi_usecase_enhancement.sql)

Create packages/db/migrations/0009_v4_multi_usecase_enhancement.sql to add the exact tables required for Use Cases 02, 04, 05, and Caregiver Continuity while maintaining 100% backward compatibility and PostgreSQL Row-Level Security (RLS):

```
-- ============================================================================
-- 0009_v4_multi_usecase_enhancement.sql
-- Multi-Track Chronic Care (UC02), OPD Slot & Queue Flow (UC04),
-- Institutional Guideline RAG & Productivity Studio (UC05), and Caregivers
-- ============================================================================

SET ROLE oncobrief_migrator;

-- 1. Extend patient with care_track, last_consultation_date, and caregiver metadata
ALTER TABLE patient
  ADD COLUMN IF NOT EXISTS care_track text NOT NULL DEFAULT 'oncology'
    CHECK (care_track IN ('oncology', 'maternal_health', 'diabetes')),
  ADD COLUMN IF NOT EXISTS last_consultation_date date,
  ADD COLUMN IF NOT EXISTS primary_language text NOT NULL DEFAULT 'en-IN';

CREATE TABLE IF NOT EXISTS caregiver_contact (
  id                  uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id              uuid NOT NULL REFERENCES organization(id),
  patient_id          uuid NOT NULL REFERENCES patient(id),
  full_name           text NOT NULL,
  relationship        text NOT NULL,
  phone_number        text NOT NULL,
  preferred_locale    text NOT NULL DEFAULT 'hi-IN',
  prefers_voice_note  boolean NOT NULL DEFAULT true,
  consent_recorded_at timestamptz NOT NULL DEFAULT now(),
  consent_recorded_by uuid NOT NULL REFERENCES app_user(id),
  is_active           boolean NOT NULL DEFAULT true,
  UNIQUE (org_id, id)
);
CREATE INDEX IF NOT EXISTS caregiver_patient_idx ON caregiver_contact (patient_id);

-- 2. Use Case 04: OPD Clinic Slots, Queue Tokens & Readiness-Gated Waitlist
CREATE TYPE opd_slot_status AS ENUM (
  'scheduled',
  'checked_in',
  'in_consult',
  'delayed_investigation',
  'completed',
  'cancelled',
  'no_show',
  'walk_in',
  'vacant'
);

CREATE TYPE admin_priority_tag AS ENUM (
  'routine_followup',
  'new_referral',
  'post_op_wound_check_documented',
  'chemo_cycle_clearance_documented',
  'high_risk_anc_milestone_documented',
  'expedited_slot_requested'
);

CREATE TABLE IF NOT EXISTS opd_appointment_slot (
  id                      uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id                  uuid NOT NULL REFERENCES organization(id),
  clinic_date             date NOT NULL DEFAULT CURRENT_DATE,
  slot_Start_time         time NOT NULL,
  slot_end_time           time NOT NULL,
  room_label              text NOT NULL DEFAULT 'OPD Room 1',
  clinician_id            uuid REFERENCES app_user(id),
  patient_id              uuid REFERENCES patient(id),
  token_number            text,
  status                  opd_slot_status NOT NULL DEFAULT 'scheduled',
  priority_tag            admin_priority_tag NOT NULL DEFAULT 'routine_followup',
  priority_source_fact_id uuid REFERENCES evidence_fact(id),
  bottleneck_note         text,
  checked_in_at           timestamptz,
  updated_by              uuid REFERENCES app_user(id),
  updated_at              timestamptz NOT NULL DEFAULT now(),
  UNIQUE (org_id, id)
```

```
);
CREATE INDEX IF NOT EXISTS opd_slot_date_idx ON opd_appointment_slot (org_id, clinic_date, slot_start_time);

CREATE TABLE IF NOT EXISTS opd_waitlist_entry (
  id                     uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id                 uuid NOT NULL REFERENCES organization(id),
  patient_id             uuid NOT NULL REFERENCES patient(id),
  care_track             text NOT NULL DEFAULT 'oncology',
  requested_date         date NOT NULL DEFAULT CURRENT_DATE,
  priority_tag           admin_priority_tag NOT NULL DEFAULT 'routine_followup',
  priority_source_fact_id uuid REFERENCES evidence_fact(id),
  status                 text NOT NULL DEFAULT 'waiting'
                           CHECK (status IN ('waiting', 'offered', 'slotted', 'cancelled')),
  offered_slot_id        uuid REFERENCES opd_appointment_slot(id),
  created_at             timestamptz NOT NULL DEFAULT now(),
  UNIQUE (org_id, id)
);

-- 3. Use Case 05: Span-Anchored Institutional SOPs & Society Guidelines (NCG, FOGSI, RSSDI)
CREATE TABLE IF NOT EXISTS guideline_document (
  id               uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id           uuid NOT NULL REFERENCES organization(id),
  code             text NOT NULL,
  title            text NOT NULL,
  issuing_body     text NOT NULL, -- e.g. 'National Cancer Grid (NCG)', 'FOGSI', 'RSSDI'
  care_track       text NOT NULL CHECK (care_track IN ('oncology', 'maternal_health', 'diabetes')),
  version_label    text NOT NULL,
  publication_date date NOT NULL,
  content_sha256   bytea NOT NULL,
  is_active        boolean NOT NULL DEFAULT true,
  uploaded_by      uuid NOT NULL REFERENCES app_user(id),
  uploaded_at      timestamptz NOT NULL DEFAULT now(),
  UNIQUE (org_id, code, version_label)
);

CREATE TABLE IF NOT EXISTS guideline_span (
  id                   uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id               uuid NOT NULL REFERENCES organization(id),
  guideline_document_id uuid NOT NULL REFERENCES guideline_document(id) ON DELETE CASCADE,
  page_number          integer NOT NULL CHECK (page_number >= 1),
  section_heading      text NOT NULL,
  span_index           integer NOT NULL,
  verbatim_text        text NOT NULL,
  bbox_x               real NOT NULL DEFAULT 0.05,
  bbox_y               real NOT NULL DEFAULT 0.10,
  bbox_w               real NOT NULL DEFAULT 0.90,
  bbox_h               real NOT NULL DEFAULT 0.08,
  keywords             text[] NOT NULL DEFAULT '{}',
  UNIQUE (guideline_document_id, page_number, span_index)
);

-- 4. Use Case 05: Clinician-Controlled Professional Content & Academic Studio Drafts
CREATE TYPE studio_artifact_kind AS ENUM (
  'referral_letter',
  'tumor_board_deck',
  'academic_case_abstract',
  'patient_info_sheet'
);

CREATE TABLE IF NOT EXISTS studio_artifact (
  id                  uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id              uuid NOT NULL REFERENCES organization(id),
  patient_id          uuid NOT NULL REFERENCES patient(id),
  artifact_kind       studio_artifact_kind NOT NULL,
  title               text NOT NULL,
  locale              text NOT NULL DEFAULT 'en-IN',
  markdown_body       text NOT NULL,
  cited_fact_ids      uuid[] NOT NULL DEFAULT '{}',
  cited_guideline_ids uuid[] NOT NULL DEFAULT '{}',
  is_deidentified     boolean NOT NULL DEFAULT false,
  status              packet_status NOT NULL DEFAULT 'draft',
  snapshot_sha256     bytea,
  created_by          uuid NOT NULL REFERENCES app_user(id),
  created_at          timestamptz NOT NULL DEFAULT now(),
  approved_by         uuid REFERENCES app_user(id),
  approved_at         timestamptz,
  UNIQUE (org_id, id)
);
```

```
-- 5. Enable Forced Row-Level Security (RLS) & Grants on all new tables
ALTER TABLE caregiver_contact ENABLE ROW LEVEL SECURITY;
ALTER TABLE caregiver_contact FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON caregiver_contact
  USING (org_id = current_setting('app.org_id', true)::uuid);

ALTER TABLE opd_appointment_slot ENABLE ROW LEVEL SECURITY;
ALTER TABLE opd_appointment_slot FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON opd_appointment_slot
  USING (org_id = current_setting('app.org_id', true)::uuid);

ALTER TABLE opd_waitlist_entry ENABLE ROW LEVEL SECURITY;
ALTER TABLE opd_waitlist_entry FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON opd_waitlist_entry
  USING (org_id = current_setting('app.org_id', true)::uuid);

ALTER TABLE guideline_document ENABLE ROW LEVEL SECURITY;
ALTER TABLE guideline_document FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON guideline_document
  USING (org_id = current_setting('app.org_id', true)::uuid);

ALTER TABLE guideline_span ENABLE ROW LEVEL SECURITY;
ALTER TABLE guideline_span FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON guideline_span
  USING (org_id = current_setting('app.org_id', true)::uuid);

ALTER TABLE studio_artifact ENABLE ROW LEVEL SECURITY;
ALTER TABLE studio_artifact FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON studio_artifact
  USING (org_id = current_setting('app.org_id', true)::uuid);

GRANT SELECT, INSERT, UPDATE ON caregiver_contact, opd_appointment_slot, opd_waitlist_entry, studio_artifact TO
oncobrief_app;
GRANT SELECT, INSERT ON guideline_document, guideline_span TO oncobrief_app;

RESET ROLE;
```

7. Step-by-Step Implementation Plan (File-by-File Tasks)

Phase 1: Foundation -- Database Migration, Domain Vocabularies & Longitudinal Comparators (Use Case 02 Core)

Task 1.1: Apply 0009_v4_multi_usecase_enhancement.sql Migration

* Files to Create/Modify:
  * Create packages/db/migrations/0009_v4_multi_usecase_enhancement.sql
  * Modify packages/db/src/seed.ts (add new tables to TRUNCATE_TABLES)
* Verification Command: pnpm db:migrate && pnpm --filter @oncobrief/db test

Task 1.2: Expand FACT_TYPES for Pending Investigations, Instructions & Chronic Care Vital Logs

* Files to Modify:
  * packages/domain/src/vocab/fact-types.ts
  * packages/domain/src/vocab/fact-value.ts
  * packages/domain/src/packet/assemble.ts
* Implementation Details:
  1. Add 'pending_investigation.recorded', 'instruction.recorded', and 'vital_sign.as_written' to FACT_TYPES.
  2. Update slotQualifier(factType, value, observedOn?) in packages/domain/src/vocab/fact-types.ts and COMPARATORS['lab_result.as_written'] / COMPARATORS['medication.recorded'] in packages/domain/src/conflict/comparators.ts:
     * When a.observedOn and b.observedOn are both present and differ by > 1 day, COMPARATORS['lab_result.as_written'] and COMPARATORS['medication.recorded'] return 'agree' (indicating a valid longitudinal time series rather than a cross-document contradiction).
     * When a.observedOn and b.observedOn are on the same date (or both undated) and values disagree, return 'disagree' so genuine same-day lab/medication discrepancies still enter the Reconciliation Room.
* Verification Command: pnpm --filter @oncobrief/domain test

Task 1.3: Build Longitudinal Lab Flowsheet & "Changes Since Last Visit" Delta Pure Domain Module

* Files to Create:
  * packages/domain/src/journey/longitudinal-review.ts
  * packages/domain/test/longitudinal-review.test.ts
* Exported Signatures:

```
export interface IntervalDeltaSummary {
  baselineDate: string | null;
  newDocumentsCount: number;
  newFacts: TimelineFactInput[];
  reopenedOrNewConflictsCount: number;
  pendingInvestigations: TimelineFactInput[];
  documentedInstructions: TimelineFactInput[];
}

export interface LabSeriesRow {
  testName: string;
  unitText: string | null;
  points: {
    factId: string;
    observedOn: string;
    valueText: string;
    numericValue: number | null;
    state: EvidenceStateValue;
    documentId: string;
    documentName: string;
    pageNumber: number | null;
  }[];
  deltaText: string | null; // Purely arithmetic difference e.g. "-1.2 g/dL since 02 Jun 2025", zero clinical
interpretation
}

export function computeIntervalDelta(
  facts: readonly TimelineFactInput[],
  documents: readonly DocumentMetaInput[],
  baselineDate: string | null,
): IntervalDeltaSummary;

export function buildLongitudinalLabFlowsheet(
  facts: readonly TimelineFactInput[],
): LabSeriesRow[];
```

* Verification Command: pnpm --filter @oncobrief/domain test -- longitudinal-review.test.ts

Task 1.4: Enhance RuleBasedExtractor & SarvamVisionOcrAdapter + Fix Upload OCR Routing

* Files to Modify:
  * packages/adapters/src/rule-based-extractor.ts
  * packages/adapters/src/sarvam-vision-ocr.ts
  * apps/web/src/lib/actions.ts
* Implementation Details:
  1. In rule-based-extractor.ts:
     * Add regex rules for pending_investigation.recorded (matching /(?:pending|awaiting|advised|scheduled for|to undergo)\s+(IHC|immunohistochemistry|PET-CT|MRI|CT scan|biopsy|OGTT|HbA1c|retinal exam|ultrasound|mammography|blood investigation[^.;\n]{0,40})/i).
     * Add regex rules for instruction.recorded (matching /(?:please bring|bring|return with|carry)\s+([^.\n]{5,90})/i).
     * Add chronic-care medications (Metformin, Glimepiride, Empagliflozin, Insulin Glargine, Folic Acid, Iron Sucrose, Calcium, Labetalol, Nifedipine) and procedures (TIFFA scan, Antenatal ultrasound, Fundus examination, Foot neuropathy screening, Oral glucose tolerance test).
  2. In sarvam-vision-ocr.ts:
     * Fix splitPdfIntoBatches so it preserves real PDF page streams instead of overwriting >10-page PDFs with %BATCH placeholder text, and accept a dynamic language parameter (hi-IN, ta-IN, te-IN, etc.).
  3. In apps/web/src/lib/actions.ts (ocrPort and extractorPort, lines 313-320):
     * Route dynamically based on env.ocrDriver (SarvamVisionOcrAdapter when SARVAM_API_KEY is configured, TextractOcrAdapter when AWS Textract is configured, and LocalMultimodalOcrAdapter as fallback).
* Verification Command: pnpm --filter @oncobrief/adapters test

---

Phase 2: Use Case 02 UI Upgrades -- Interval Delta Banner, Lab Flowsheets, Batch Verify &

Multi-Track Seed

Task 2.1: Seed Multi-Track Patients (DEMO-MAT-004 & DEMO-DIA-005), OPD Slots & Guideline Library

* Files to Modify:
  * packages/db/src/fixtures.ts
  * packages/db/src/seed.ts
* Implementation Details:
  1. Add DEMO-MAT-004 ("Priya Nair -- High-Risk ANC Trimester 2 (FOGSI Track)") with serial Hemoglobin (11.2 g/dL -> 9.8 g/dL), BP logs, OGTT report, and a missing required TIFFA Anomaly Scan report.
  2. Add DEMO-DIA-005 ("Rajesh Iyer -- Long-Term Type 2 Diabetes Follow-Up (RSSDI Track)") with 3 serial HbA1c reports (8.9% -> 7.8% -> 7.1%), Serum Creatinine/eGFR panel, Metformin + Empagliflozin prescriptions, and a missing required Annual Dilated Retinal Fundus Exam.
  3. Seed last_consultation_date on DEMO-001, DEMO-MAT-004, and DEMO-DIA-005 so the "Changes Since Last Visit" delta banner is populated out of the box.
* Verification Command: pnpm demo:reset

Task 2.2: Add verifyEvidenceBatchAction & Guided "Next Step" Readiness Bar on /patients/[id]/evidence

* Files to Modify:
  * packages/db/src/services/ledger.ts (export verifyEvidenceBatch)
  * apps/web/src/lib/actions.ts (export verifyEvidenceBatchAction)
  * apps/web/src/app/(app)/patients/[id]/evidence/page.tsx
  * apps/web/src/components/TimelineNodeView.tsx
* Implementation Details:
  1. Add a top Consultation Readiness & Changes Since Last Visit Banner on evidence/page.tsx showing:
     * Current ReadinessBand + exact next actions to reach ready (e.g., 1 Conflict to Resolve -> link to Reconciliation Room; 1 Required Document Missing -> link to Record Map; Batch Verify High-Confidence Facts).
     * A Since Last Visit filter chip alongside All facts, Unverified, Conflicting, Verified, and Lab & Vitals Flowsheet.
  2. Remove the hardcoded 'Apollo Hospitals' and 'CT Scan (Thorax)' strings from TimelineNodeView.tsx, replacing them with row.issuingFacility ?? 'Facility not stated' and accurate modality labels for Oncology, Maternal Health, and Diabetes.
* Verification Command: pnpm --filter @oncobrief/web test

Task 2.3: Wire Real ABDM v6.5.0 & HL7 mCODE STU4 Bundle Import and Document Linking in DocumentUploadForm.tsx

* Files to Modify:
  * apps/web/src/lib/actions.ts (add importStandardBundleAction and linkExistingDocumentAction)
  * apps/web/src/components/DocumentUploadForm.tsx
* Implementation Details:
  * Replace the two alert(...) buttons in DocumentUploadForm.tsx (lines 329-344) with real <form action={importStandardBundleAction}> buttons that construct and ingest a full NRCeS ABDM v6.5.0 OPConsultRecord / DiagnosticReportRecord bundle or HL7 mCODE STU4 bundle into the patient's Evidence Ledger.
* Verification Command: pnpm --filter @oncobrief/web test

Phase 3: Use Case 04 -- Smart OPD Clinic Flow, Slot Backfill & Care-Team Task Assignment

Task 3.1: Build OPD Flow & Readiness-Gated Slot Backfill Service (packages/db/src/services/opd-flow.ts)

* Files to Create:
  * packages/domain/src/flow/slot-optimizer.ts
  * packages/domain/test/slot-optimizer.test.ts
  * packages/db/src/services/opd-flow.ts
* Exported Signatures (slot-optimizer.ts):

```
export interface BackfillCandidate {
  waitlistEntryId: string;
  patientId: string;
  patientName: string;
  careTrack: 'oncology' | 'maternal_health' | 'diabetes';
  readinessBand: ReadinessBand;
  priorityTag: string;
  prioritySourceQuote: string | null;
```

```
  matchScore: number;
  rationale: string; // e.g. "Record READY (0 gaps, 0 conflicts) - Documented post-op check"
}

export function rankWaitlistForVacantSlot(
  vacantSlot: OpdSlotInput,
  candidates: readonly WaitlistCandidateInput[],
): BackfillCandidate[];
```

* Verification Command: pnpm --filter @oncobrief/domain test -- slot-optimizer.test.ts

Task 3.2: Upgrade /workspace with Live OPD Clinic Flow Board, Track Filter & Pilot Telemetry Bar

* Files to Modify:
  * apps/web/src/app/(app)/workspace/page.tsx
  * apps/web/src/lib/actions.ts (add updateOpdSlotStatusAction, backfillOpdSlotAction, registerWalkInAction)
* Implementation Details:
  1. Add a 60-90 Day Pilot Operational Telemetry Bar at the top of /workspace displaying live database metrics:
     * Consult-Ready Rate (ready / total scheduled)
     * Span-Validation Gate Pass Rate (promoted / total candidates)
     * Source Contradictions Caught Pre-Consult
     * OPD Slot Utilization & Backfill Count
  2. Add a view toggle on /workspace: Readiness Worklist View vs. Live OPD Clinic Flow & Slot Board (showing time slots 09:00-12:30, Token #, Arrival/Check-In status, Delayed Lab blocker badges, and a 1-click "Backfill Vacant/Blocked Slot from Ready Waitlist" action).
  3. Add Care Track filter pills (All Tracks, Oncology (NCG), Maternal Health (FOGSI), Diabetes (RSSDI)).
* Verification Command: pnpm --filter @oncobrief/web test

Task 3.3: Add In-Row Task Assignee Selector & Gap Waiver UI

* Files to Modify:
  * apps/web/src/app/(app)/patients/[id]/tasks/page.tsx
  * apps/web/src/app/(app)/patients/[id]/record-map/page.tsx
* Implementation Details:
  1. Query organization staff members (listOrgMembers(q, orgId)) and render an inline <form action={assignTaskAction}> dropdown on every task card in tasks/page.tsx so tasks can be assigned to Dr. Meera Rao (clinician), Anita Joseph (coordinator), or Deepak Nair (records_officer).
  2. Add a <details> waiver form on record-map/page.tsx wired to waiveGapAction so authorized staff (checklist:author) can waive non-applicable expected items with a mandatory audited reason.
* Verification Command: pnpm --filter @oncobrief/web test

Phase 4: Use Case 05 -- Doctor Productivity Studio & Span-Grounded Guideline Co-Pilot

Task 4.1: Build Institutional Guideline & SOP Retrieval Engine ([G#] Citations)

* Files to Modify:
  * apps/web/src/lib/assistant/classify.ts
  * apps/web/src/lib/assistant/tools.ts
  * apps/web/src/lib/assistant/prompt.ts
  * apps/web/src/lib/assistant/orchestrator.ts
* Implementation Details:
  1. In classify.ts: Add 'guideline_knowledge_question' intent. Route queries asking about NCG, FOGSI, RSSDI, institutional SOPs, standard workup checklists, or reference protocols to 'guideline_knowledge_question', while continuing to block patient-specific diagnostic/prescribing demands ("what dose should I prescribe for this patient" -> 'unsupported_medical_question').
  2. In tools.ts:
     * Fix the bug on line 301 where openTasksCount was computed from timeline instead of listTasks(q, patientId).
     * Add searchGuidelines(q, query, careTrack?) querying guideline_span + guideline_document and returning verbatim protocol spans with [G1], [G2] citations and bounding boxes.
  3. In orchestrator.ts and prompt.ts: Format [G#] guideline cards alongside [E#] patient evidence cards so clinicians can verify exact protocol passages with one click.
* Verification Command: pnpm --filter @oncobrief/web test

Task 4.2: Build the Clinician Productivity & Content Studio (/patients/[id]/packet & studio.ts)

* Files to Create/Modify:
  * Create packages/domain/src/studio/draft-generator.ts
  * Create packages/domain/test/draft-generator.test.ts
  * Modify apps/web/src/app/(app)/patients/[id]/packet/page.tsx
  * Modify apps/web/src/lib/actions.ts (add createStudioArtifactAction, updateStudioArtifactAction, approveStudioArtifactAction)
* Implementation Details:
  * In draft-generator.ts, implement deterministic, source-cited template compilers (with optional LLM phrasing refinement constrained to [E#]/[G#] tags) for 4 deliverables:
    1. Specialist Referral / Transfer Letter (structured clinical history as written, procedures, serial lab table, open tasks, and referring doctor sign-off).
    2. Multidisciplinary Tumor Board / Case Presentation (slide-by-slide Markdown deck: Slide 1 Case Identity & Readiness, Slide 2 Chronological Interventions [E#], Slide 3 Pathology & Imaging Spans [E#], Slide 4 Reconciled Contradictions, Slide 5 Reference NCG Guideline Checklist [G#]).
    3. De-Identified Academic Case Report Draft (automatically redacts Patient Name, MRN, ABHA, and exact dates into relative day offsets Day 0, Day +14, Day +45 for research manuscripts and teaching).
    4. Patient & Caregiver Visit Summary Handout (plain-language administrative summary of verified appointments, documents to bring, and hospital contact numbers).
  * Add a Print / Save as PDF view (@media print stylesheet) alongside Export ABDM v6.5.0 FHIR JSON and Export HL7 mCODE STU4 JSON buttons on packet/page.tsx.
* Verification Command: pnpm --filter @oncobrief/domain test -- draft-generator.test.ts

---

Phase 5: Design Principles -- Caregiver Voice Network, Accessibility (WCAG 2.2 AA) & Mobile Remediation

Task 5.1: Upgrade /patients/[id]/continuity with Caregiver Registry & 11-Language Spoken Audio Preview

* Files to Modify:
  * apps/web/src/app/(app)/patients/[id]/continuity/page.tsx
  * apps/web/src/components/ContinuityAudioPlayer.tsx (new client component)
  * packages/db/src/seed.ts (seed templates across en, hi, ta, te, bn, mr, kn, ml, gu, pa, or)
* Implementation Details:
  1. Render the Consented Family Caregiver Card on continuity/page.tsx allowing coordinators to add/view a caregiver (fullName, relationship, phoneNumber, preferredLocale) and toggle Sync reminders to family caregiver.
  2. Add a "Listen / Preview Voice Note (Sarvam Bulbul v3)" button on every composed or approved patient message that calls /api/v1/assistant/voice/tts with the message's bodyRendered and locale (falling back gracefully to browser window.speechSynthesis if offline) so low-literacy voice outreach is demonstrable live.
* Verification Command: pnpm --filter @oncobrief/web test

Task 5.2: Remediate WCAG 2.2 AA Contrast, Form Labels, Landmarks & 375px Mobile Overflow

* Files to Modify:
  * apps/web/src/app/globals.css
  * apps/web/src/app/(app)/layout.tsx
  * apps/web/src/app/(app)/patients/[id]/layout.tsx
  * apps/web/src/components/PatientTabs.tsx
  * apps/web/src/components/AssistantLauncher.tsx
* Implementation Details (accessibility-inclusive-design skill compliance):
  1. Contrast Fixes (globals.css):
     * Darken --color-state-extracted from #9a6b12 (4.26:1, fails AA) to #7c530b (6.12:1 on #f7f4ee, passes WCAG AA).
     * Darken --color-rule-strong from #b9b0a0 (1.96:1, fails WCAG 1.4.11) to #857d6f (3.45:1 on #f7f4ee, passes 3:1 UI boundary contrast).
  2. 375px Mobile Overflow Fixes:
     * Add flex-wrap to the authenticated header in apps/web/src/app/(app)/layout.tsx:12.
     * Wrap PatientTabs in an overflow-x-auto container with aria-label="Patient sections".
     * Add bottom padding (pb-20) to patient pages so the fixed Ask OncoBrief launcher (AssistantLauncher.tsx) never obscures Source B or bottom evidence rows.

3. Semantic Landmarks & Live Regions:
   * Change the wrapper in apps/web/src/app/(app)/patients/[id]/layout.tsx:37 to <main id="main-content"> and add a keyboard-accessible "Skip to main content" link in (app)/layout.tsx.
   * Add an aria-live="polite" status region for verification, conflict resolution, and task updates.
4. Audit Redaction Hardening (packages/db/src/services/audit.ts):
   * Add 'reason' and 'filename' to REDACT_KEYS or hash free-text reasons so PHI can never leak into audit_event.metadata_json.

* Verification Command: pnpm --filter @oncobrief/web test:e2e

8. Risk & Dependency Matrix

Risk ID	Risk Description	Prob.	Impact	Concrete Mitigation Strategy	Contingency / Fallback Trigger
R-01	Clinical Boundary Violation: Adding guideline RAG (UC 05) or OPD priority tags (UC 04) trips check-clinical-boundary.ts or appears diagnostic to judges.	Med	High	Keep all guideline retrieval strictly as verbatim source-span lookup ([G#]) with a mandatory non-clinical header; restrict OPD priority tags to documented administrative categories (post_op_check_documented, chemo_clearance_documented) anchored to a source span.	Run pnpm boundary:check on every commit; any forbidden token fails CI immediately.
R-02	External API Unavailability During Live Judging: SARVAM_API_KEY, NVIDIA_API_KEY, or AWS Bedrock credentials hit rate limits (429) or network timeouts during the demo.	Med	High	Maintain deterministic offline fallbacks across every port: LocalMultimodalOcrAdapter for PDFs/FHIR/DICOM, RuleBasedExtractor for facts/instructions, synthesizeDeterministicGroundedAnswer for chat/guidelines, and Web Speech API / pre-generated WAV fallback for TTS audio preview.	Automatic fallback on timeout (>8s) or missing API key with zero UI crash.
R-03	False Suppression of Real Same-Day Lab Conflicts: Relaxing the lab comparator to allow longitudinal trends accidentally hides two conflicting lab reports from the same day.	Low	High	Scope COMPARATORS['lab_result.as_written'] strictly by date distance: only treat differing values as a longitudinal series when abs(dateA - dateB) >= 1 day; same-day or undated disagreements still trigger conflicting.	Unit test both same-day conflict (DOM-LAB-CONFLICT) and multi-day series (DOM-LAB-SERIES) in conflict.test.ts.

R-04	Clinician Rubber-Stamping in Batch Verification: A user clicks "Batch Verify" on low-quality OCR spans without reviewing them.	Med	Med	Restrict verifyEvidenceBatchAction strictly to facts where confidence_band === 'high' (ocr_confidence >= 0.85), state === 'extracted', and conflict_count === 0. Low-confidence (needs_review) and conflicting facts are excluded from batch verify and require individual inspection.	Modal lists every fact and quote included in the batch before commit.

9. Master Verification & Testing Matrix

Every phase is gated by automated tests across all 5 architectural layers:

Layer	Test Suite / Command	New Verification Scenarios Added in V4	Pass Criteria
Layer 1: Database & RLS	pnpm --filter @oncobrief/db test	Tenant isolation on caregiver_contact, opd_appointment_slot, opd_waitlist_entry, guideline_document, studio_artifact; immutability of text_span and ledger_entry.	100% pass, 0 cross-tenant leaks
Layer 2: Pure Domain	pnpm --filter @oncobrief/domain test	longitudinal-review.test.ts (interval deltas & serial lab flowsheets); slot-optimizer.test.ts (readiness-gated slot backfill ranking); draft-generator.test.ts (de-identified case report & referral letter citations); check-clinical-boundary.ts.	100% pass, 0 boundary violations
Layer 3: Multi-Modal Adapters	pnpm --filter @oncobrief/adapters test	Pending investigation & instruction extraction in rule-based-extractor; multi-page PDF batching & Indic language codes in sarvam-vision-ocr; ABDM v6.5.0 & mCODE round-trip ingestion.	100% pass
Layer 4: Web & Assistant	pnpm --filter @oncobrief/web test	[G#] guideline retrieval grounding; batch verification RBAC & audit logging; OPD slot status mutations & task assignment actions.	100% pass
Layer 5: Browser E2E & Mobile	pnpm --filter @oncobrief/web test:e2e	Full walkthrough of UC 02 (Interval Delta + Lab Flowsheet + Batch Verify), UC 04 (OPD Slot Board + Ready Waitlist Backfill), UC 05 (Guideline Query + Referral/Tumor Board Studio), Caregiver Voice TTS preview, and scrollWidth <= 375 mobile check.	100% E2E pass at 1440px and 375px